

MUTUAL FUND SIP ANALYSIS

End-to-End Data Analytics Project using Python, MySQL & Power BI

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Tools Used: Python | Pandas | MySQL | Power BI | GitHub

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1. Project Overview

Introduction

Mutual funds have become one of the most preferred investment options for individuals seeking long-term wealth creation. Among various investment methods, Systematic Investment Plans (SIPs) provide investors with a disciplined and consistent approach to investing. As the number of investors and transactions increases, financial institutions generate large volumes of data that can be analyzed to understand investor behavior, fund performance, and investment trends.

This project presents an end-to-end data analytics solution using Python, MySQL, and Power BI. The workflow begins with data cleaning and preprocessing in Python, followed by exploratory data analysis (EDA), relational database creation and business query analysis using MySQL, and finally the development of an interactive Power BI dashboard to visualize key insights.

2. Business Problem

Financial institutions require data-driven insights to improve investment strategies and customer satisfaction. This project addresses the following business questions:

- Which mutual funds receive the highest SIP investments?
- Which investor segments contribute the highest investments?
- Which cities and states have the highest investment activity?
- How does investment behavior vary across different risk profiles?
- What trends can be observed in monthly SIP investments?
- Which mutual funds have the highest expected returns and average NAV?

Answering these questions helps organizations understand investor behavior and make informed business decisions.

3. Project Objectives

The primary objectives of this project are:

- Clean and preprocess raw mutual fund datasets.
- Perform Exploratory Data Analysis (EDA) to understand investor demographics and investment patterns.
- Design a relational database using MySQL.

- Execute business-oriented SQL queries to solve real-world analytical problems.
- Develop an interactive Power BI dashboard.
- Generate meaningful business insights to support decision-making.

4. Dataset Description

The dataset consists of three related tables.

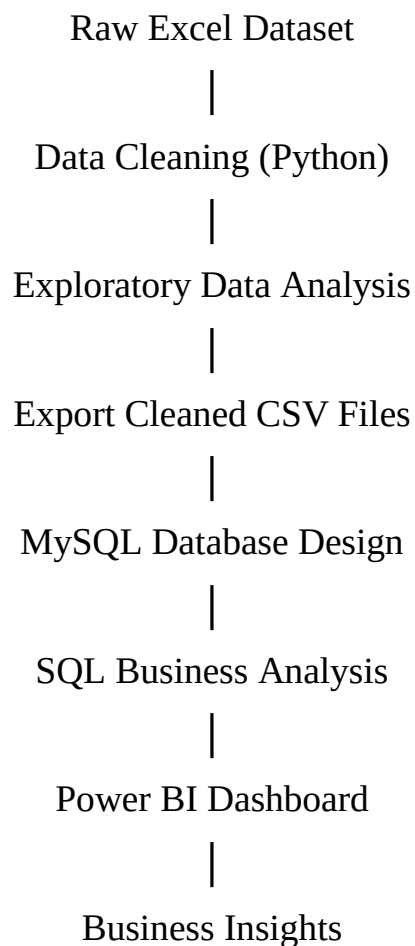
Table Name	Description
Investors	Investor demographic information including age, gender, occupation, income, city, state, risk profile, and investment goals.
Fund_Master	Mutual fund information including fund category, AMC, and expected returns.
SIP_Transactions	Monthly SIP transaction details including SIP amount, NAV, and units allocated.

Dataset Summary

Dataset	Number of Records
Investors	1,000
Fund Master	5
SIP Transactions	12,000

5. Methodology

The project follows an end-to-end data analytics workflow.



Data Cleaning Steps

The following preprocessing tasks were performed:

- Imported data from Excel into Python.
- Checked data types for each column.
- Verified missing values.
- Removed duplicate records.
- Validated numerical fields such as SIP Amount and NAV.
- Standardized column names and formats.
- Exported cleaned datasets as CSV files for SQL and Power BI analysis.

6. Exploratory Data Analysis (Python)

Exploratory Data Analysis was performed to understand investor demographics and investment characteristics before building the dashboard.

Key Analyses

Age Distribution of Investors

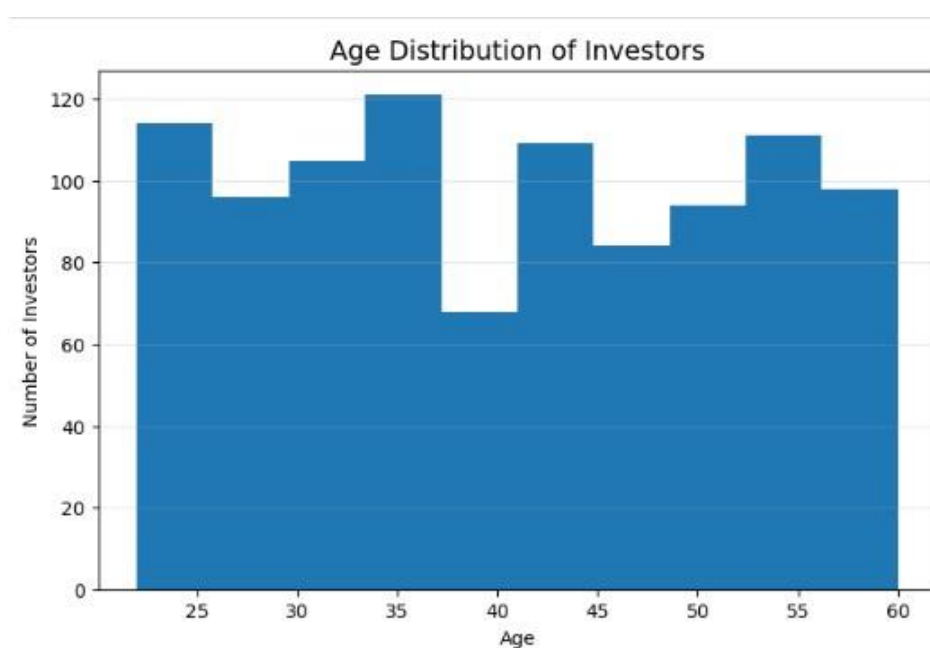
Gender Distribution

Risk Profile Distribution

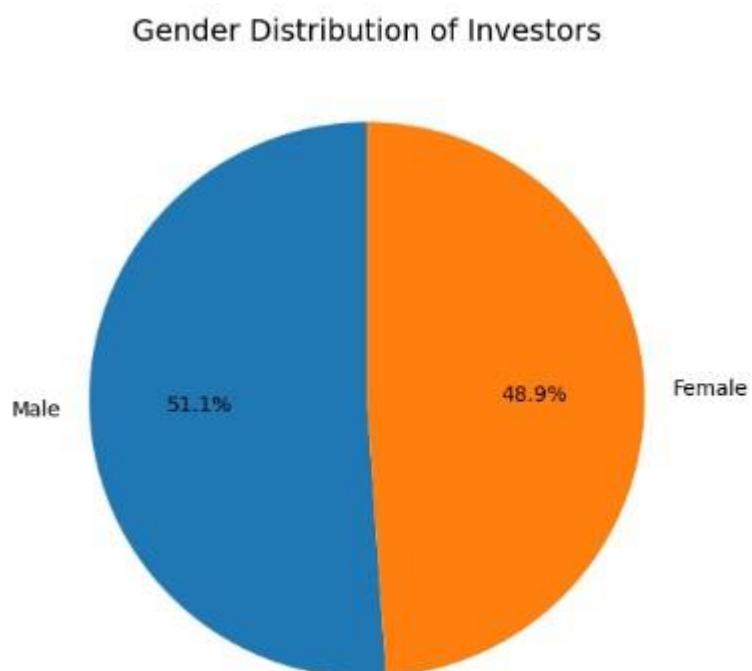
Average Annual Income by Occupation

charts

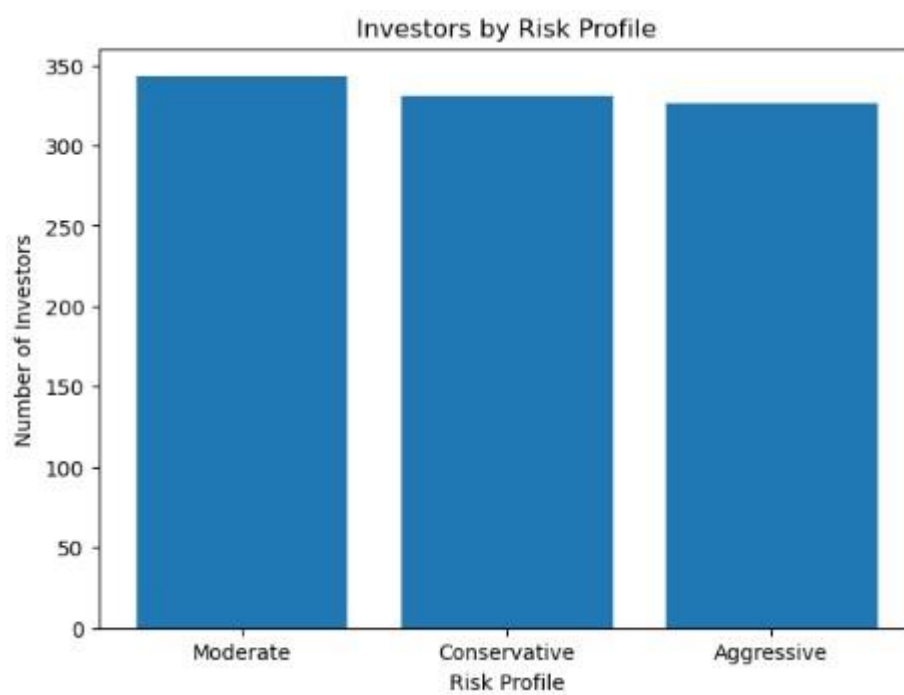
 Age Distribution



Gender Distribution



Risk Profile Distribution



Observations

The investor base primarily consists of middle-aged individuals.

The distribution of investors across genders provides insight into customer demographics.

Moderate-risk investors represent a significant portion of the investor population.

7. SQL Database Design & Analysis

A relational database named mutual_fund_db was created in MySQL. Three tables were designed with appropriate primary and foreign key relationships.

Tables Created

- Investors
- Fund_Master
- SIP_Transactions

SQL Concepts Used

1. SELECT
2. WHERE
3. GROUP BY
4. ORDER BY
5. INNER JOIN
6. HAVING
7. Aggregate Functions
8. Views

A total of 32 business-oriented SQL queries were executed to analyze investor behavior, fund performance, and investment trends.

SQL Query 1: Database Schema

Table: investors

Columns:

<u>Investor_ID</u>	varchar(20) PK
Gender	varchar(20)
Age	int
Marital_Status	varchar(30)
Occupation	varchar(100)
Annual_Income	decimal(15,2)
City	varchar(100)
State	varchar(100)
Risk_Profile	varchar(30)
Investment_Goal	varchar(100)
Investment_Horizon_Years	int
Investor_Segment	varchar(50)

Table: fund_master

Columns:

<u>Fund_ID</u>	varchar(20) PK
Fund_Name	varchar(100)
Fund_Category	varchar(100)
AMC	varchar(100)
Expected_Return	decimal(5,2)

Table: sip_transactions

Columns:

<u>Transaction_ID</u>	varchar(20) PK
<u>Investor_ID</u>	varchar(20)
Date	date
<u>Fund_ID</u>	varchar(20)
Fund_Name	varchar(100)
SIP_Amount	decimal(10,2)
NAV	decimal(10,2)
Units_Allocated	decimal(12,4)

Explanation:

The database consists of three related tables: Investors, Fund_Master, and SIP_Transactions. The Investor_ID and Fund_ID columns establish relationships between the tables using primary and foreign keys, ensuring data integrity and enabling efficient relational analysis.

SQL Query 2: Investor and Fund Details Using JOIN

Business Question: Which investor has invested in which mutual fund?

SQL Concepts Used:

- INNER JOIN
- SELECT

```
-- 11) Which investor invested in which fund?
• SELECT
    i.Investor_ID,
    i.City,
    i.Risk_Profile,
    s.Fund_Name,
    s.SIP_Amount
FROM Investors i
JOIN SIP_Transactions s
ON i.Investor_ID = s.Investor_ID;
```


Explanation:

This query combines the Investors and SIP_Transactions tables using the Investor_ID field. It displays each investor along with the mutual fund they invested in and the corresponding SIP amount. This helps analyze investment patterns at the individual investor level.

SQL Query 3: Fund-wise Total Investment

Business Question: Which mutual fund has received the highest total SIP investment?

SQL Concepts Used:

- GROUP BY
- SUM()
- ORDER BY

```
-- 12) Which mutual fund received the highest investment?  
• SELECT  
    Fund_Name,  
    SUM(SIP_Amount) AS Total_Investment  
FROM SIP_Transactions  
GROUP BY Fund_Name  
ORDER BY Total_Investment DESC;
```

Explanation:

This query calculates the total SIP investment received by each mutual fund by summing all SIP amounts. The results are sorted in descending order to identify the funds attracting the highest investments.

SQL Query 4: Average SIP by Risk Profile

Business Question: Which risk profile contributes the highest average SIP amount?

SQL Concepts Used:

- INNER JOIN
- GROUP BY
- AVG()

```
-- 21) Which risk profile contributes the highest average SIP amount?
SELECT
    i.Risk_Profile,
    ROUND(AVG(s.SIP_Amount),2) AS Average_SIP
FROM Investors i
JOIN SIP_Transactions s
ON i.Investor_ID = s.Investor_ID
GROUP BY i.Risk_Profile
ORDER BY Average_SIP DESC;
```

Explanation:

This query joins the Investors and SIP_Transactions tables to calculate the average SIP amount for each risk profile. It helps identify which category of investors contributes higher average investments.

SQL Query 5: High-Value Investors (HAVING)

Business Question: Which investors have invested more than ₹1,50,000?

SQL Concepts Used:

- GROUP BY
- HAVING
- SUM()

```
-- 23) Investors with Total Investment > ₹1,50,000 (HAVING)
SELECT
    Investor_ID,
    SUM(SIP_Amount) AS Total_Investment
FROM SIP_Transactions
GROUP BY Investor_ID
HAVING SUM(SIP_Amount) > 150000
ORDER BY Total_Investment DESC;
```

Explanation:

This query groups transactions by investor and calculates the total investment made by each investor. The HAVING clause filters only those investors whose total investment exceeds ₹1,50,000, helping identify high-value investors.

8. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize investor behaviour, fund performance, and SIP investment trends. The dashboard consists of four pages, each focusing on a different aspect of the analysis.

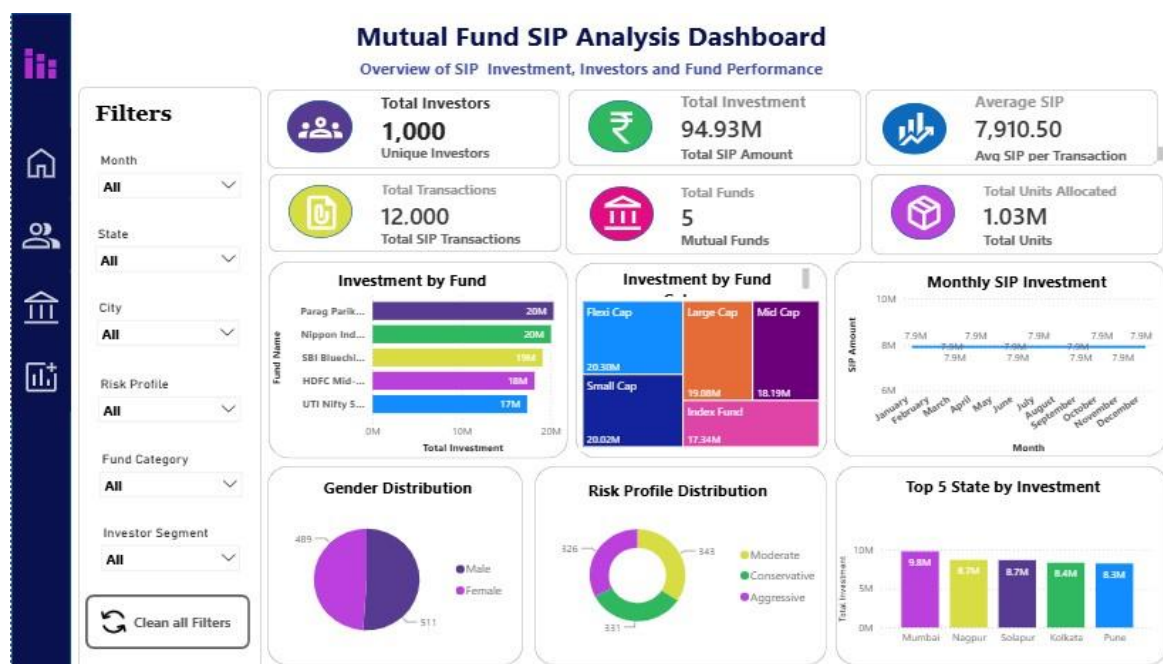
8.1 Executive Dashboard

Objective

The Executive Dashboard provides a high-level overview of the mutual fund SIP portfolio through key performance indicators (KPIs) and summary visuals. It enables quick monitoring of overall investment performance and supports strategic decision-making.

Key Performance Indicators (KPIs)

- Total Investors
- Total Investment
- Average SIP Amount
- Total Mutual Funds
- Total Transactions
- Maximum SIP Amount
- Minimum SIP Amount
- Total Units Allocated



Explanation

The Executive Dashboard summarizes the overall performance of the mutual fund portfolio using KPI cards and visualizations. It provides decision-makers with a quick overview of the investor base, total investment value, transaction volume, and SIP statistics. Interactive slicers allow users to filter the data dynamically for deeper analysis.

8.2 Investor Analysis Dashboard

Objective

This dashboard focuses on analyzing investor demographics and investment behavior across different customer segments.

Visuals Included

- Investors by Age Group
- Gender Distribution
- Occupation-wise Investors
- Investor Segments
- State-wise Investors
- Risk Profile Distribution



Explanation

The Investor Analysis Dashboard helps identify demographic patterns among investors. It enables the comparison of investor distribution by age, gender, occupation, state, and risk profile, providing valuable insights into customer characteristics and investment preferences.

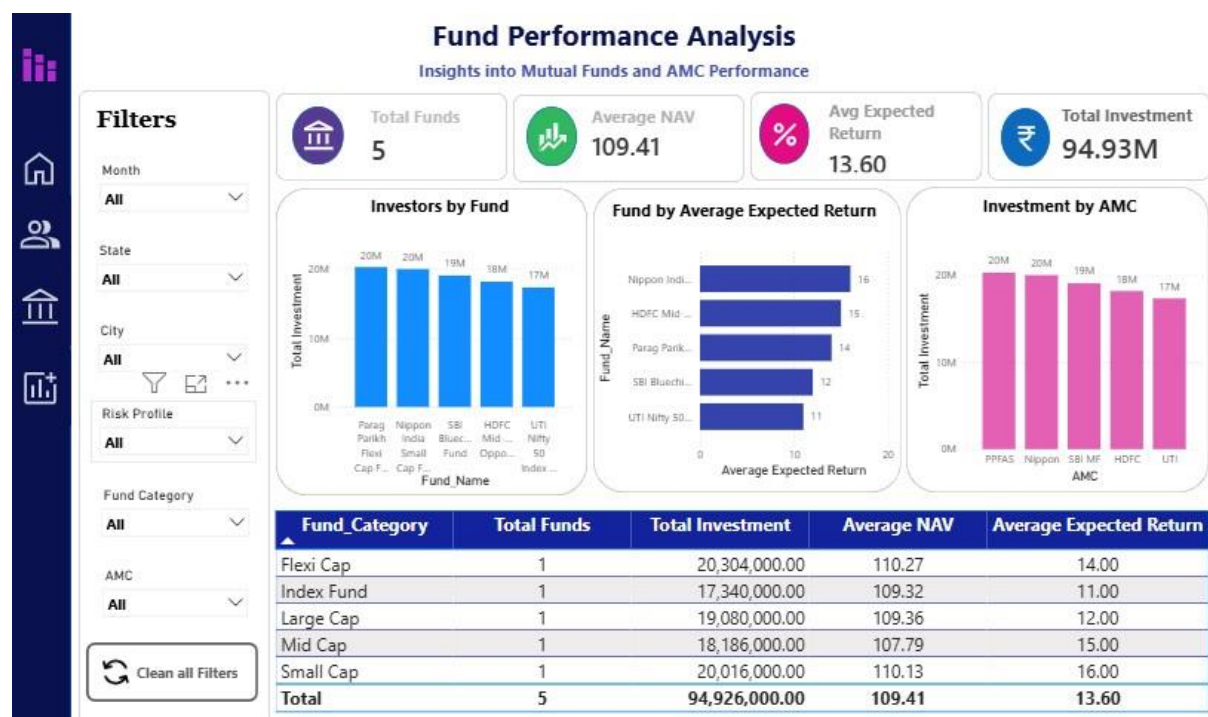
8.3 Fund Performance Dashboard

Objective

This dashboard evaluates the performance of mutual funds based on investment distribution and fund-related metrics.

Visuals Included

- Investment by Fund
- Investment by AMC
- Fund Category
- Expected Return
- Average NAV



Explanation

The Fund Performance Dashboard compares mutual funds based on total investments, expected returns, and average NAV values. It assists in identifying

high-performing funds and understanding investment preferences across different Asset Management Companies (AMCs).

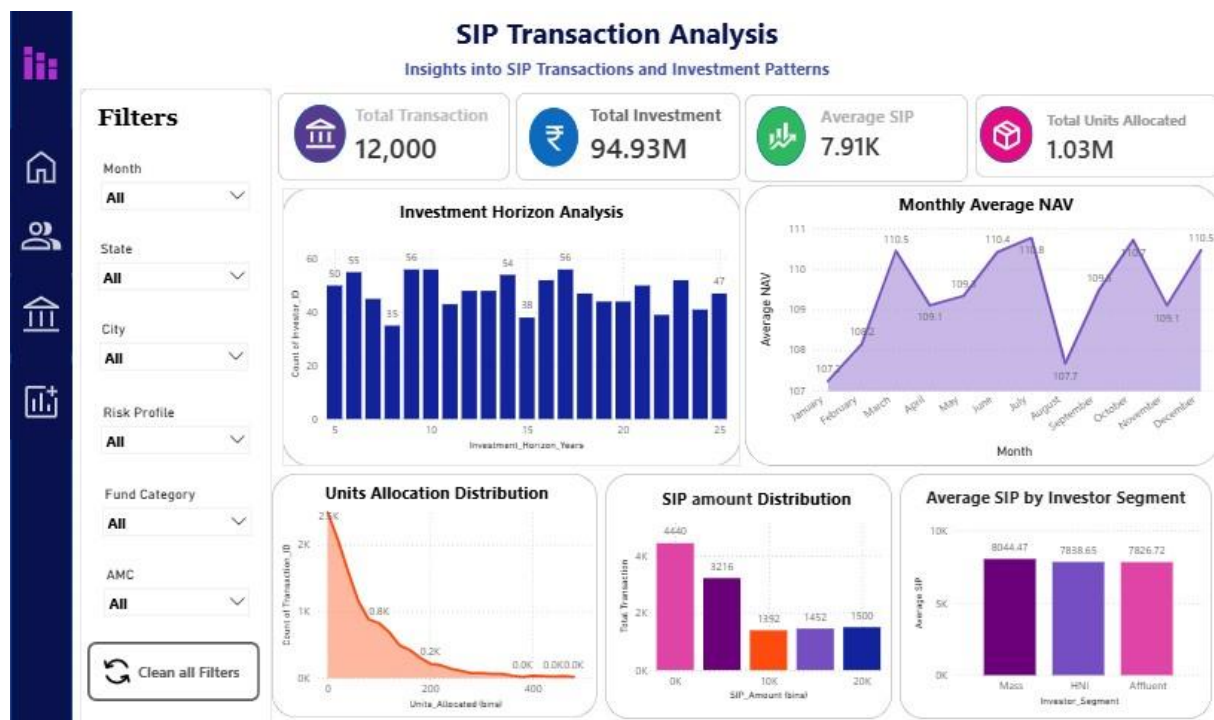
8.4 SIP Transaction Dashboard

Objective

This dashboard analyzes SIP transaction trends and investment patterns over time.

Visuals Included

- Monthly SIP Trend
- State-wise Investment
- Investment by Risk Profile
- Fund-wise Investment
- Units Allocated



Explanation

The SIP Transaction Dashboard provides insights into monthly investment trends and transaction behavior. It highlights investment distribution across states, risk profiles, and mutual funds while also displaying the total units allocated through SIP investments.

Dashboard Features

The Power BI dashboard includes the following features:

- Interactive slicers for dynamic filtering.
- KPI cards for quick performance monitoring.
- Bar, column, pie, donut, line, and treemap visualizations.
- DAX measures for calculating key business metrics.
- Navigation buttons for seamless page transitions.

9. Business Insights

The analysis of the mutual fund SIP dataset using Python, MySQL, and Power BI provided the following business insights:

1. Strong Investor Base

The dashboard shows a total of **1,000 investors** who completed **12,000 SIP transactions**, contributing a **total investment of ₹94.93 million**. This indicates a healthy investor base with consistent participation in SIP investments.

2. Average SIP Contribution

The average SIP amount is ₹7,910.50 per transaction. This reflects the average monthly contribution made by investors and indicates a stable investment pattern suitable for long-term wealth creation.

3. Fund-wise Investment Analysis

Among the five mutual funds analyzed, Parag Parikh Flexi Cap Fund and Nippon India Small Cap Fund received the highest total investment of approximately ₹20 million each. On the other hand, UTI Nifty 50 Index Fund recorded the lowest investment at approximately ₹17 million.

Business Value: Investors appear to prefer Flexi Cap and Small Cap funds, indicating greater interest in diversified and high-growth investment opportunities.

4. Fund Category Performance

The Flexi Cap category attracted the highest total investment of ₹20.30 million, followed closely by the Small Cap category with ₹20.02 million. The Index Fund category recorded the lowest investment of ₹17.34 million.

Business Value: Investors show stronger preference for actively managed equity-oriented funds compared with passive index funds.

5. Expected Return Analysis

The Nippon India Small Cap Fund offers the highest expected return of 16%, while the UTI Nifty 50 Index Fund has the lowest expected return of 11%. Across all funds, the average expected return is 13.60%.

Business Value: Funds with higher expected returns are likely to attract investors seeking higher long-term growth, although investment decisions should also consider the associated level of risk.

6. Investor Demographics

The average investor age is 40.68 years, indicating that the investor base mainly consists of middle-aged individuals who are likely investing for long-term financial goals such as retirement planning, children's education, or wealth accumulation.

7. Gender Distribution

The investor base is almost evenly distributed, with 511 male investors and 489 female investors.

Business Value: Since participation is balanced across genders, financial institutions can design investment products and awareness campaigns that effectively address both male and female investors.

8. Age Group Distribution

The 46–60 years age group has the highest number of investors (130), followed by the 36–45 years group (127). The number of investors is comparatively lower among younger age groups.

Business Value: Marketing strategies can be developed to encourage greater participation from younger investors while continuing to serve the needs of experienced investors.

9. Occupation-wise Investment

The Salaried segment contributed the highest total investment of approximately ₹19.4 million, followed closely by Government Employees and Self-Employed individuals.

Business Value: Salaried professionals represent the largest contributor to SIP investments, making them an important target segment for mutual fund companies.

10. Risk Profile Analysis

Among all investor categories, the Moderate Risk profile has the highest number of investors (343), followed by Conservative (331) and Aggressive (326) investors.

Business Value: The majority of investors prefer balanced investment options that provide reasonable returns while controlling investment risk.

11. State-wise Investment

The analysis indicates that Mumbai recorded the highest total investment of approximately ₹9.8 million, followed by Nagpur (₹8.7 million), Solapur (₹8.7 million), Kolkata (₹8.4 million), and Pune (₹8.3 million).

Business Value: These locations represent important markets where financial institutions can strengthen customer engagement and expand investment services.

12. Investment Horizon

Most investors have an investment horizon ranging between 8 and 20 years, indicating a long-term investment approach through SIPs.

Business Value: Long-term investors generally contribute to stable and recurring investment inflows, benefiting both investors and fund management companies.

13. NAV Trend

The monthly Average NAV remained within the range of 107.7 to 110.5, showing relatively stable movement throughout the year.

Business Value: Stable NAV values indicate consistent fund performance and provide confidence to long-term investors.

14. Units Allocated

The dashboard shows a total of 1.03 million units allocated, demonstrating substantial investment activity and indicating efficient conversion of SIP contributions into mutual fund units.

10. Conclusion

This project successfully demonstrated a complete data analytics workflow using Python, MySQL, and Power BI. Raw investment data was transformed into meaningful insights through data cleaning, SQL analysis, and interactive dashboard development. The project highlights how analytical techniques can support better decision-making in the financial sector by identifying investment patterns, understanding investor behavior, and evaluating fund performance.

11. Future Scope

Future improvements to this project may include:

- Predicting investor behavior using machine learning techniques.
- Forecasting future SIP investment trends.
- Integrating real-time mutual fund market data.
- Developing automated dashboards connected to live databases.
- Expanding the analysis with additional financial performance metrics.